

- **Lead through programming** - A skilled human worker guides the robot through the sequence of motions it is to execute using manual controls. This precise sequence is stored in the memory of the robot with great accuracy and can be repeatedly executed at the touch of a button thereafter.
 - One-time labour cost, high installation cost
 - Requires a human only for the initial training stage of any operation
- **Autonomous control** - The most independent control schema, this is the optimisation of input signal-processing, machine learning and objective maximisation to have control that is completely independent of a human input
 - No labour cost (excluding programming the controller), high development cost
 - Requires significant thought, software design and possible mathematical analysis

How it is controlled - more on autonomous control

Autonomous robots have the incredible burden of needing to be literally ‘unplug and play’ - they are expected to be fully capable of certain tasks from the first day of operation, with no external training. Further, they’re expected to get marginally, if not phenomenally, better at their tasks over time.

Consider the simple example of a simple floor-cleaning robot. It would be expected to figure out the best path around a room so as to cover the entire accessible floor in the smallest time. Further, it may be expected to plan such that the clean parts of the floor are not traversed again by the possibly dirt-carrying robot. It may be further expected to have a particular battery life such that it can clean the whole house with just one charge. Under autonomous control, all of these concerns would have to be appropriately addressed and parametrised into the robot’s limited language of signal input, processing and output.

The most common kinds of control ‘architecture’ used to design autonomous robots are:

- **Top-Down (the brain-first approach)** : The idea is to build the optimal brain after taking a macroscopic design and functional view of the requirements. The argument is that If the brain is built properly, it will be able to process all the expected kinds of input signals to generate useful control output. Such ‘proper’ building of the brain often entails training the processor on huge and varied datasets, comprehensive program-